



WORKSHOP MIKROTIK BANDWIDTH MANAGEMENT

**Agus Setiawan
Juliandi Pratama
Nixtrain**

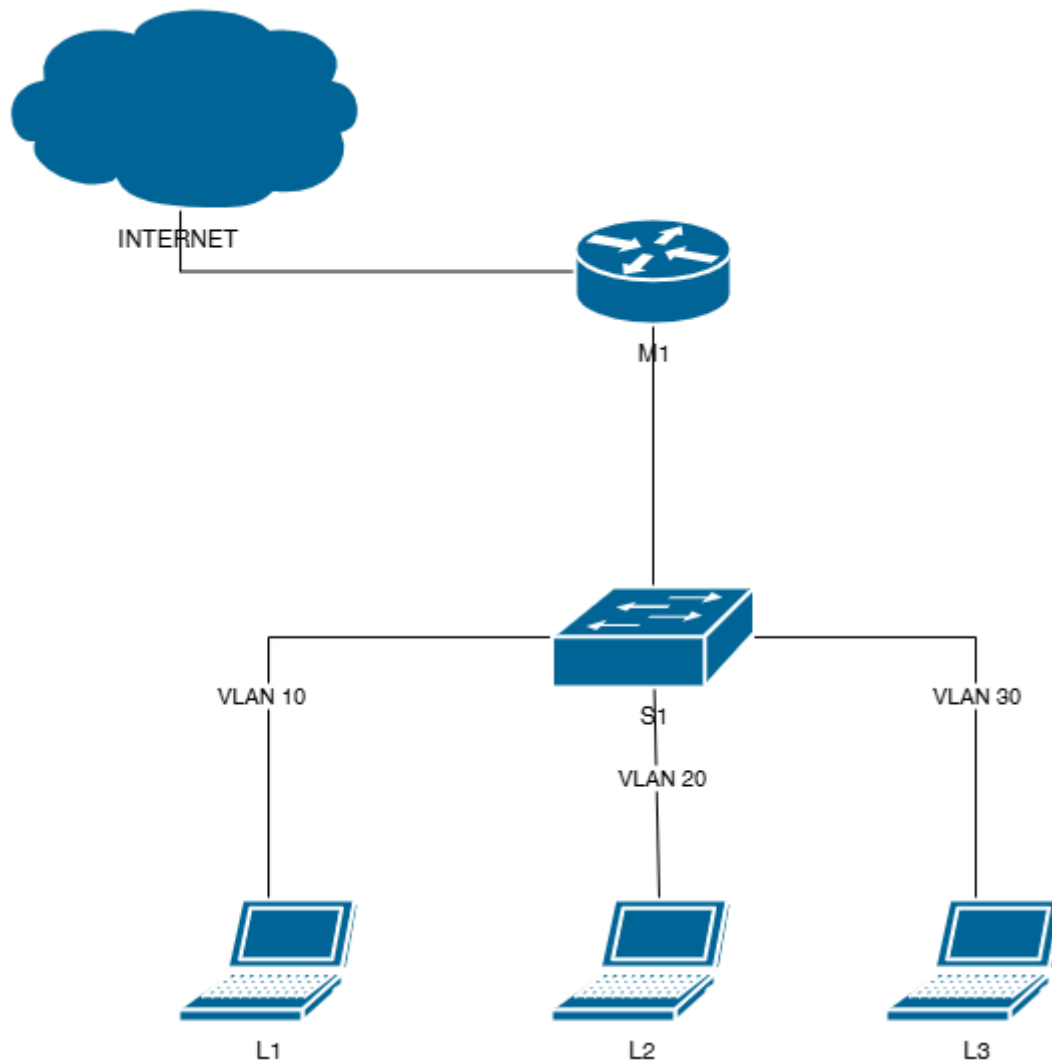
2025

Daftar Isi

Lab 1. Topologi Jaringan LAN, Device Information dan IP Addressing	2
Lab 2. Basic Internet Configuration Router	4
Lab 3. Simple Queue	11
Lab 4. PCQ (Per Connection Queue)	12
Lab 5. Queue Tree + Mangle	13

Lab 1. Topologi Jaringan LAN, Device Information dan IP Addressing

Topology



Device Information

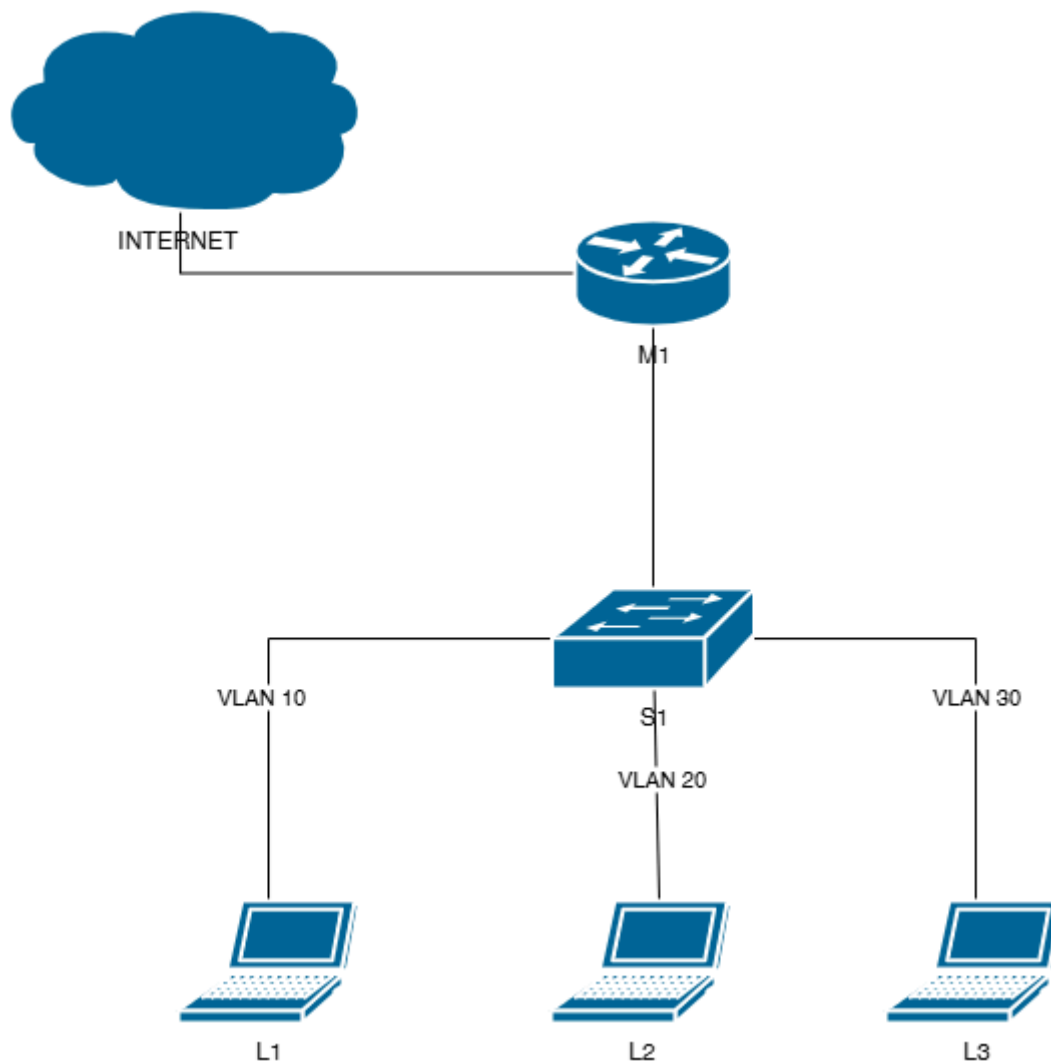
Hostname	Device Model	Software Version
Internet	N/A	N/A
M1	RB941-2nD	RouterOS Version 7.8
S1	Cisco WS-C2950-24	C2950-I6K2L2Q4-M Version 12.1(22)EA13
L1	Laptop	Windows 11
L2	Laptop	Windows 11
L3	Laptop	Windows 11

IP Addressing

Segmen	IP Addressing
Internet	DHCP
VLAN 10	172.16.10.0/24
VLAN 20	172.16.20.0/24
VLAN 30	172.16.30.0/24

Lab 2. Basic Internet Configuration Router

Topology



Konfigurasi

- Konfigurasi basic internet router mikrotik agar semua host bisa terhubung ke internet

M1

```
# sep/15/2025 11:32:13 by RouterOS 7.8
# software id = J30K-TPCV
#
# model = RB941-2nD
# serial number = D1130F6CD2A4
/interface ethernet
set [ find default-name=ether1 ] name=ether1-WAN
set [ find default-name=ether2 ] name=ether2-LAN
set [ find default-name=ether4 ] name=ether4-MANAGEMENT
/interface wireless
set [ find default-name=wlan1 ] ssid=MikroTik
/interface vlan
add interface=ether2-LAN name=vlan10 vlan-id=10
add interface=ether2-LAN name=vlan20 vlan-id=20
add interface=ether2-LAN name=vlan30 vlan-id=30
/interface wireless security-profiles
set [ find default=yes ] supplicant-identity=MikroTik
/ip pool
add name=dhcp_pool0 ranges=172.16.10.11-172.16.10.20
add name=dhcp_pool1 ranges=172.16.20.11-172.16.20.20
add name=dhcp_pool2 ranges=172.16.30.11-172.16.30.20
/ip dhcp-server
add address-pool=dhcp_pool0 interface=vlan10 name=dhcp1
add address-pool=dhcp_pool1 interface=vlan20 name=dhcp2
add address-pool=dhcp_pool2 interface=vlan30 name=dhcp3
/ip address
add address=172.16.10.1/24 interface=vlan10
network=172.16.10.0
add address=172.16.20.1/24 interface=vlan20
network=172.16.20.0
add address=172.16.30.1/24 interface=vlan30
network=172.16.30.0
add address=10.10.10.1/24 interface=ether4-MANAGEMENT
network=10.10.10.0
/ip dhcp-client
add interface=ether1-WAN
/ip dhcp-server network
add address=172.16.10.0/24 gateway=172.16.10.1
add address=172.16.20.0/24 gateway=172.16.20.1
add address=172.16.30.0/24 gateway=172.16.30.1
/ip dns
set allow-remote-requests=yes
/ip firewall nat
add action=masquerade chain=srcnat out-interface=ether1-WAN
/ip service
set ftp disabled=yes
set www disabled=yes
set api disabled=yes
set api-ssl disabled=yes
/system clock
set time-zone-name=Asia/Jakarta
```

```
/system identity
set name=M1
/system ntp client
set enabled=yes
/system ntp server
set broadcast=yes enabled=yes manycast=yes multicast=yes
/system ntp client servers
add address=id.pool.ntp.org
```

S1

```
S2#show run
Building configuration...

Current configuration : 2483 bytes
!
version 12.1
no service pad
service timestamps debug uptime
service timestamps log uptime
service password-encryption
!
hostname S2
!
enable secret 5 $1$933M$fEljP7wIAa/wDtGRPHpcC/
!
ip subnet-zero
!
!
spanning-tree mode pvst
no spanning-tree optimize bpdu transmission
spanning-tree extend system-id
!
!
!
!
interface FastEthernet0/1
  switchport mode trunk
!
interface FastEthernet0/2
  switchport access vlan 10
  switchport mode access
!
interface FastEthernet0/3
  switchport access vlan 10
  switchport mode access
!
interface FastEthernet0/4
  switchport access vlan 10
  switchport mode access
```

```
!  
interface FastEthernet0/5  
  switchport access vlan 10  
  switchport mode access  
!  
interface FastEthernet0/6  
  switchport access vlan 10  
  switchport mode access  
!  
interface FastEthernet0/7  
  switchport access vlan 10  
  switchport mode access  
!  
interface FastEthernet0/8  
  switchport access vlan 10  
  switchport mode access  
!  
interface FastEthernet0/9  
  switchport access vlan 20  
  switchport mode access  
!  
interface FastEthernet0/10  
  switchport access vlan 20  
  switchport mode access  
!  
interface FastEthernet0/11  
  switchport access vlan 20  
  switchport mode access  
!  
interface FastEthernet0/12  
  switchport access vlan 20  
  switchport mode access  
!  
interface FastEthernet0/13  
  switchport access vlan 20  
  switchport mode access  
!  
interface FastEthernet0/14  
  switchport access vlan 20  
  switchport mode access  
!  
interface FastEthernet0/15  
  switchport access vlan 20  
  switchport mode access  
!  
interface FastEthernet0/16  
  switchport access vlan 20  
  switchport mode access  
!  
interface FastEthernet0/17  
  switchport access vlan 30  
  switchport mode access
```

```

!
interface FastEthernet0/18
  switchport access vlan 30
  switchport mode access
!
interface FastEthernet0/19
  switchport access vlan 30
  switchport mode access
!
interface FastEthernet0/20
  switchport access vlan 30
  switchport mode access
!
interface FastEthernet0/21
  switchport access vlan 30
  switchport mode access
!
interface FastEthernet0/22
  switchport access vlan 30
  switchport mode access
!
interface FastEthernet0/23
  switchport access vlan 30
  switchport mode access
!
interface FastEthernet0/24
  switchport access vlan 30
  switchport mode access
!
interface Vlan1
  no ip address
  no ip route-cache
  shutdown
!
ip http server
banner motd ^C
Authorized users only
^C
!
line con 0
  password 7 01100F17580405002F
  logging synchronous
  login
line vty 0 4
  password 7 00071A150754080901
  logging synchronous
  login
line vty 5 15
  login
!
!
end

```

Verifikasi

Untuk verifikasi basic konfigurasi internet dapat melakukan ping dari host ke internet dan dapat melakukan speedtest.

L1

C:\Windows\system32\cmd.exe

Microsoft Windows [Version 10.0.22631.5909]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Management>ping google.com

Pinging forcesafesearch.google.com [216.239.38.120] with 32 bytes of data:
Reply from 216.239.38.120: bytes=32 time=20ms TTL=247
Reply from 216.239.38.120: bytes=32 time=20ms TTL=247
Reply from 216.239.38.120: bytes=32 time=28ms TTL=247
Reply from 216.239.38.120: bytes=32 time=20ms TTL=247

Ping statistics for 216.239.38.120:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 20ms, Maximum = 28ms, Average = 22ms

C:\Users\Management>

← → ↻ mikrotik.com/download ☆ 📁

 **MIKROTIK**

Home About Buy Jobs Hardware Software Support Training Account

Software Downloads Changelogs Download archive RouterOS The Dude Mobile apps Back To Home

Upgrading RouterOS

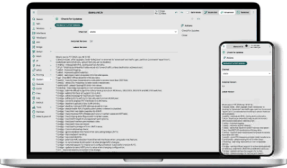
If you are already running RouterOS, upgrading to the latest version can be done by clicking on "Check For Updates" in QuickSet or System > Packages menu in WebFig or WinBox.

See the [documentation](#) for more information about upgrading and release types.

To manage your router, use the web interface, or download the maintenance utilities. Winbox to connect to your device, Dude to monitor your network and Netinstall for recovery and re-installation.

WinBox 4.0beta30

WinBox 3.42



speedtest.net/id/result/18225954869

SPEEDTEST Bahasa Indonesia Apps Learn

SEBAR ID Hasil 18225954869 HASIL PENGATURAN

UNDUHAN Mbps 34.50 UNGGAHAN Mbps 36.89

Ping ms 6 4 28

MULAI

Koneksi Tunggal

PT. Telekomunikasi Indonesia Bandung Ganti Server

Google 193.186.4.182

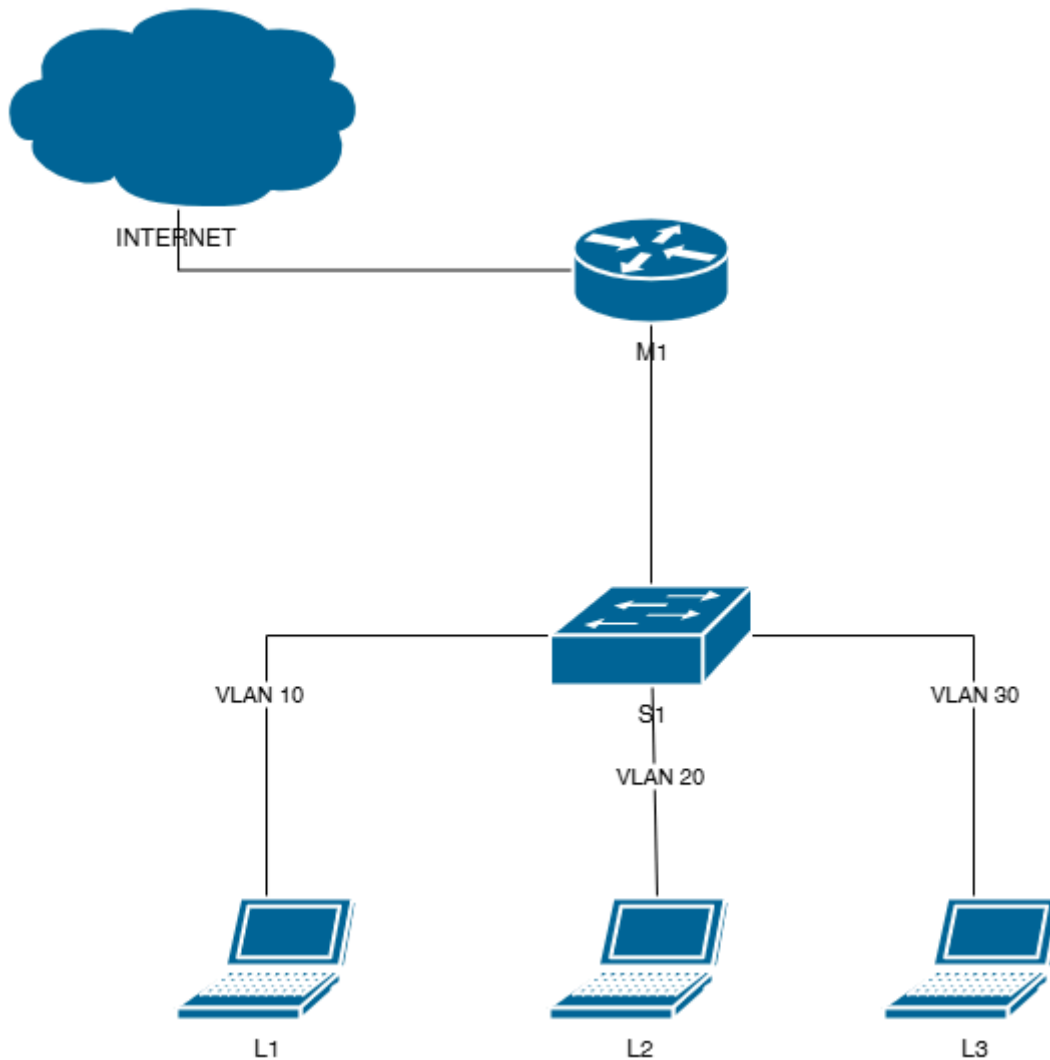
SEBERAPA BESAR KEMUNGKINAN ANDA MEREKOMENDASIKAN GOOGLE KE TEMAN ATAU KOLEGA ANDA?

0 1 2 3 4 5 6 7 8 9 10

Sama sekali tidak mungkin Sangat mungkin sekali

Lab 3. Simple Queue

Topology



Konfigurasi

- Konfigurasi bandwidth management Simple Queue

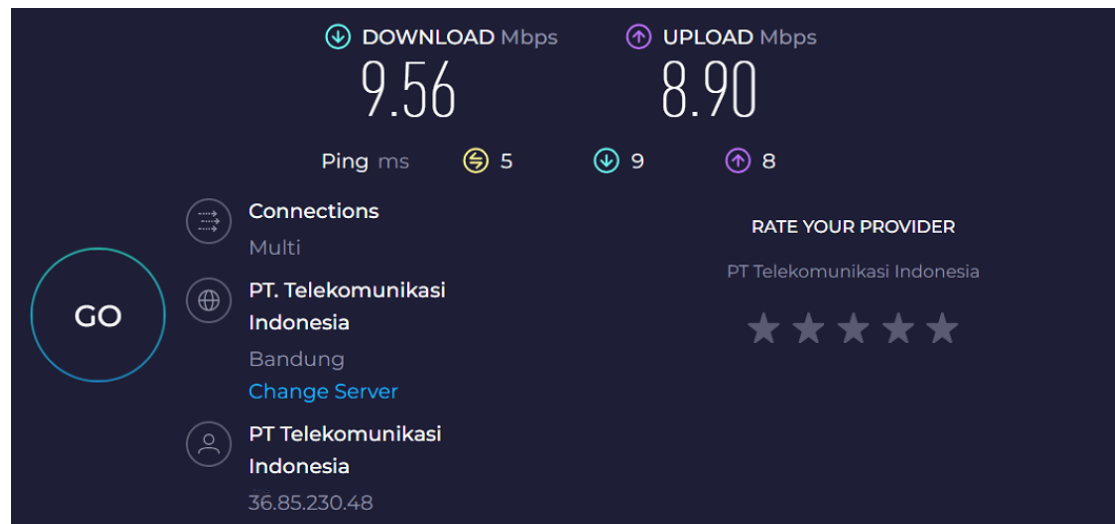
M1

```
/queue simple
add name=client_L1 target=172.16.10.0/24 max-limit=10M/10M
add name=client_L2 target=172.16.20.0/24 max-limit=20M/20M
add name=client_L3 target=172.16.30.0/24 max-limit=30M/30M
```

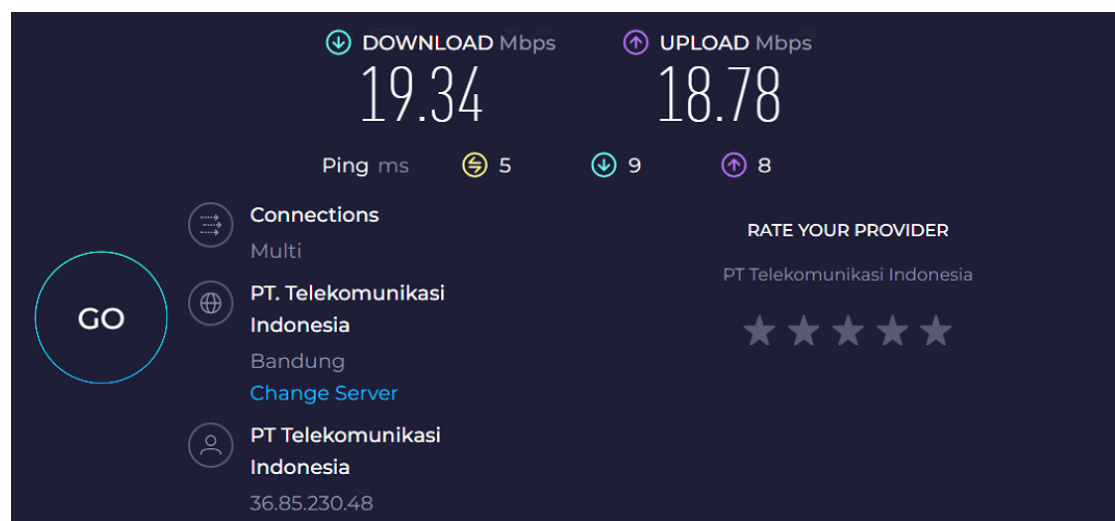
Verifikasi

Verifikasi pastikan setiap laptop sudah mendapatkan bandwidth sesuai dengan liitasi bandwidth yang diberikan.

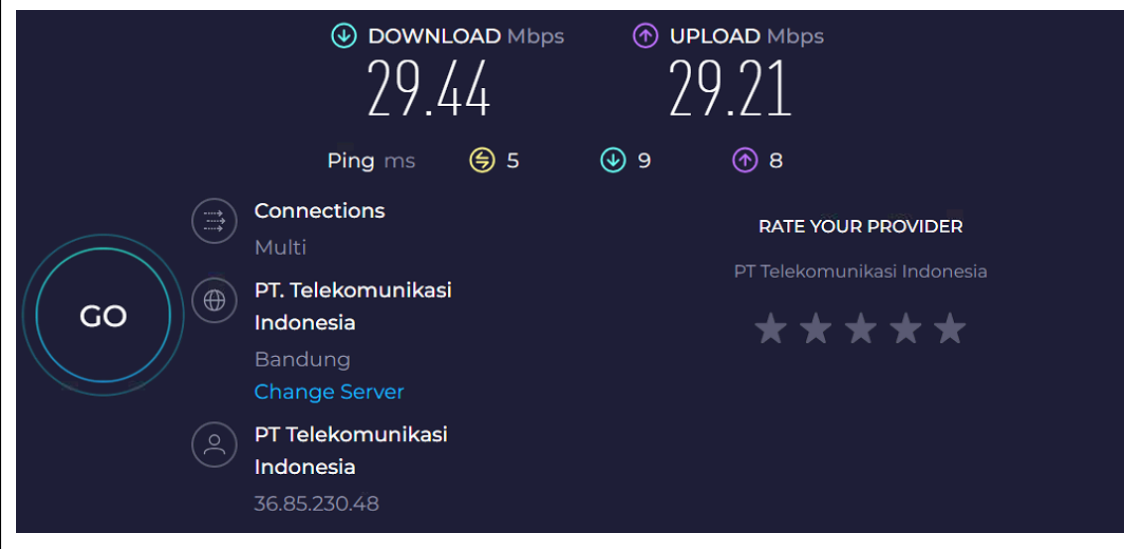
L1



L2



L3



Pada saat speedtest berjalan, maka mikrotik akan membaca bahwa adanya traffic yang masuk kedalam limitasi sehingga memberikan indikator seperti dibawah

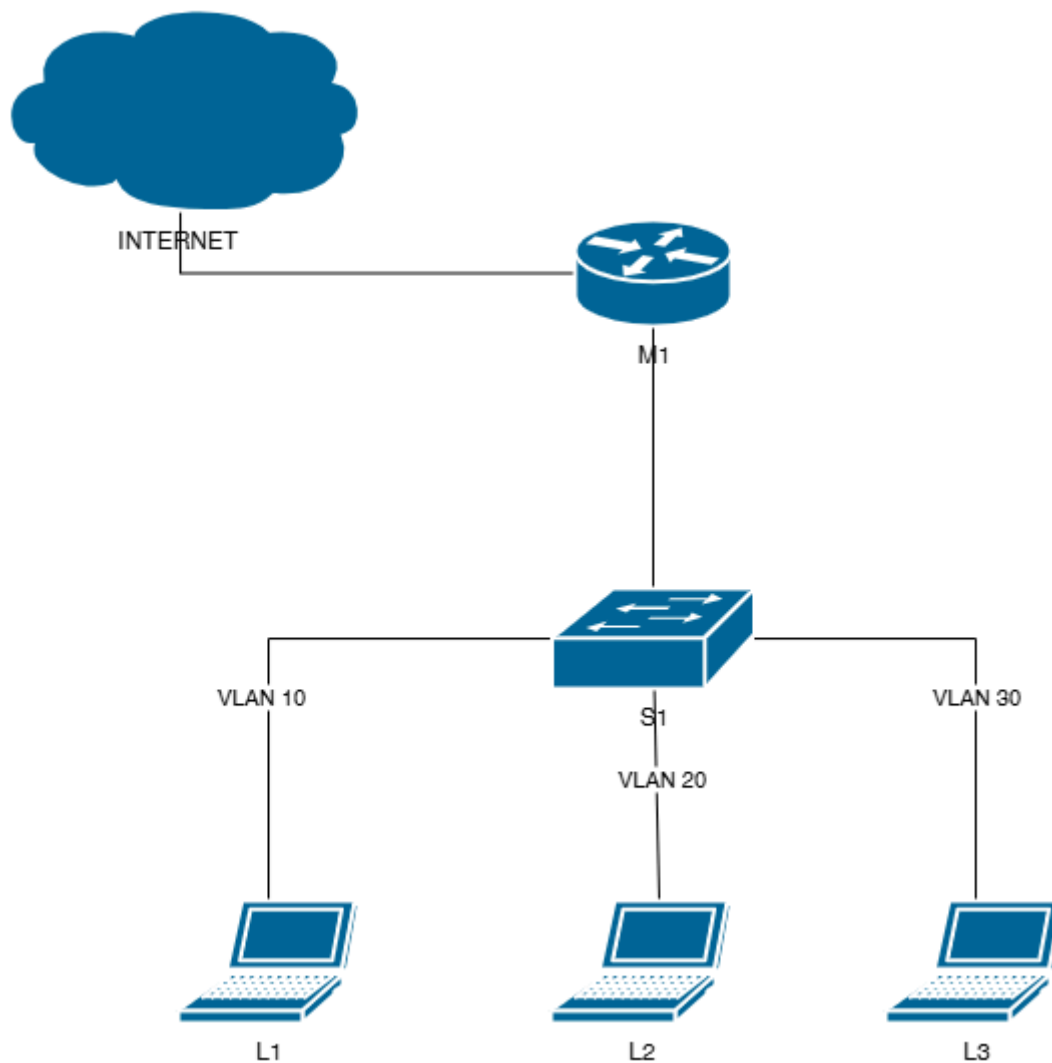
M1

Queue List									
Simple Queues Interface Queues Queue Tree Queue Types									
+ - ✓ ✗ 📁 🔍 Reset Counters Reset All Counters Find									
#	Name	Target	Upload Max Limit	Download Max Limit	Packet Marks	Upload	Download	Total Max Limit (bi...	
0	client_L1	172.16.10.0/24	10M	10M		0 bps	0 bps		
1	client_L2	172.16.20.0/24	10M	10M		859.2 kbps	9.8 Mbps		
2	client_L3	172.16.30.0/24	30M	30M		0 bps	0 bps		

3 items (1 selected) 0 B queued 0 packets queued

Lab 4. PCQ (Per Connection Queue)

Topology



Konfigurasi

Konfigurasi bandwidth management PCQ (Per Connection Queue)

M1

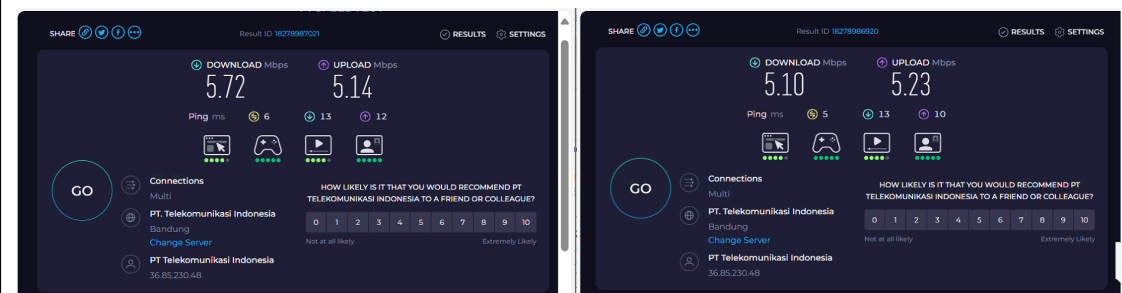
```
/queue type
add kind=pcq name=pcq-download pcq-classifier=dst-address pcq-rate=10M
add kind=pcq name=pcq-upload pcq-classifier=src-address pcq-rate=10M
```

```
/queue simple  
set client_L1 queue=pcq-upload/pcq-download
```

Verifikasi

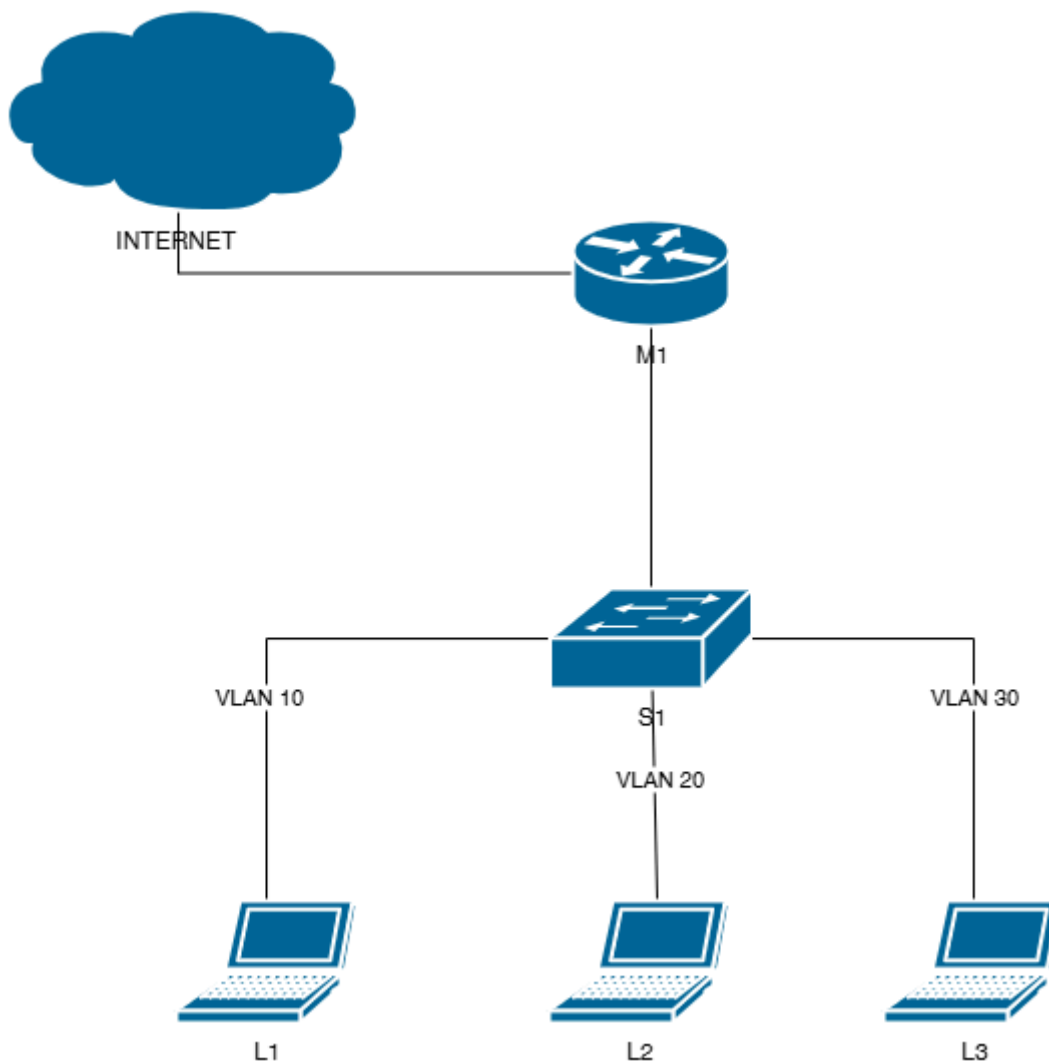
Untuk verifikasi kita harus menggunakan beberapa laptop yang terhubung ke vlan 10 dan melakukan speedtest secara bersamaan

L1



Lab 5. Queue Tree + Mangle

Topology



Konfigurasi

- Konfigurasi bandwidth management Queue Tree dan Mangle

M1

```
/ip firewall mangle
add chain=prerouting src-address=172.16.10.0/24 action=mark-packet
new-packet-mark=vlan10_up passthrough=no
```

```
add chain=postrouting dst-address=172.16.10.0/24 action=mark-packet
new-packet-mark=vlan10_down passthrough=no

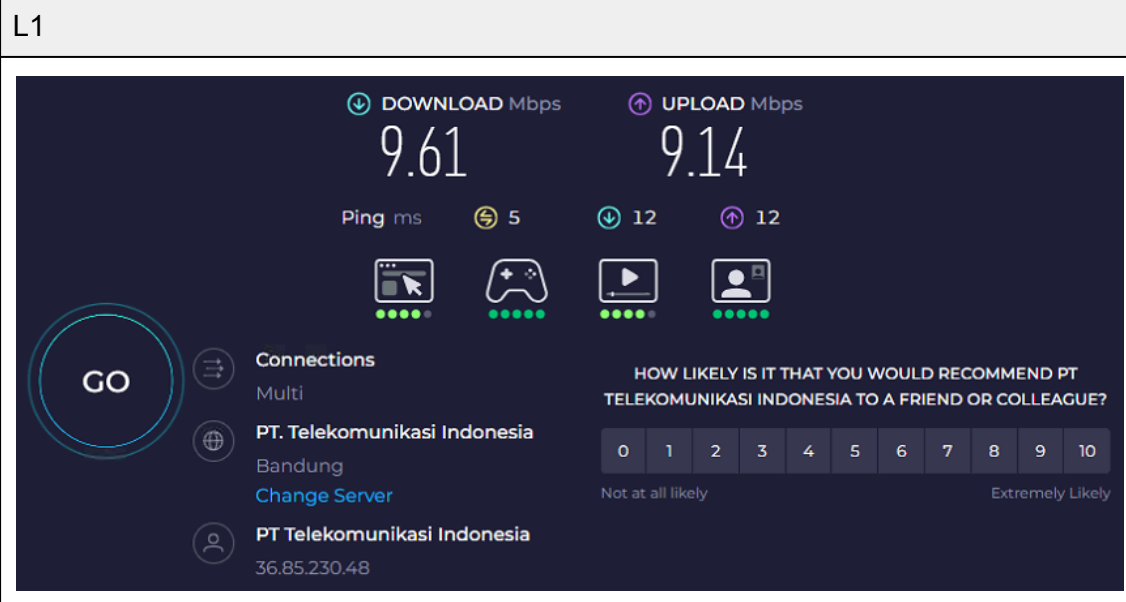
/queue tree

add name="Download VLAN10" parent=ether2-LAN packet-mark=vlan10_down
max-limit=10M

add name="Upload VLAN10" parent=ether1-WAN packet-mark=vlan10_up
max-limit=10M
```

Verifikasi

Verifikasi nya cukup lakukan speedtest lagi seperti sebelumnya dan hasilnya akan muncul sesuai dengan limitasi bandwidth yang dikonfigurasi



M1

Queue List

Name	Parent	Packet Marks	Limit At (b...	Max Limit ...	Avg. Rate	Queued By...	Bytes	Packets
Upload VLAN10	ether1-WAN	vlan10_up		10M	719.4 kbps	0 B	3025.4 ...	0
Download VLAN10	ether2-LAN	vlan10_down		10M	9.9 Mbps	8.8 KiB	16.1 MiB	0

2 items | 8.8 KiB queued | 6 packets queued

Lab 6. Backup & Restore Configuration

Konfigurasi

- Konfigurasi ini tidak akan merubah topologi dan bandwidth apapun, kita hanya akan melakukan pencadangan pada konfigurasi yang sudah terpasang sekarang untuk digunakan di masa depan apabila akan ada pergantian router
- Binary Backup (untuk restore di router sama):

M1

```
/system backup save name=backup_router
```

- Export Configuration (untuk dokumentasi/migrasi):

M1

```
/export file=export_router
```

Verifikasi

Cek apakah file backup atau export sudah ada di folder file

M1

```
/file print
```

Restore

- Binary Restore Konfigurasi

M1

```
/system backup load name=backup_router.backup
```

- Import Restore Konfigurasi

M1

```
/import file=export_router.rsc
```